

Can SEAS5 seasonal forecasts drive daily streamflow prediction? Machine learning experiments in Central Asian catchments

Study Region and Methods

Study Region

- Central Asia:
 - 3 nival–glacial mountain catchments
 - 4 rainfed steppe catchments
- Europe:
 - 2 Alpine catchments for validation



Datasets

- ECMWF SEAS5 seasonal forecasts (1981-2025)
- 7 variables: temperature (mean, min, max), precipitation, snowfall, snow depth, solar radiation
- Lead times up to 214 days

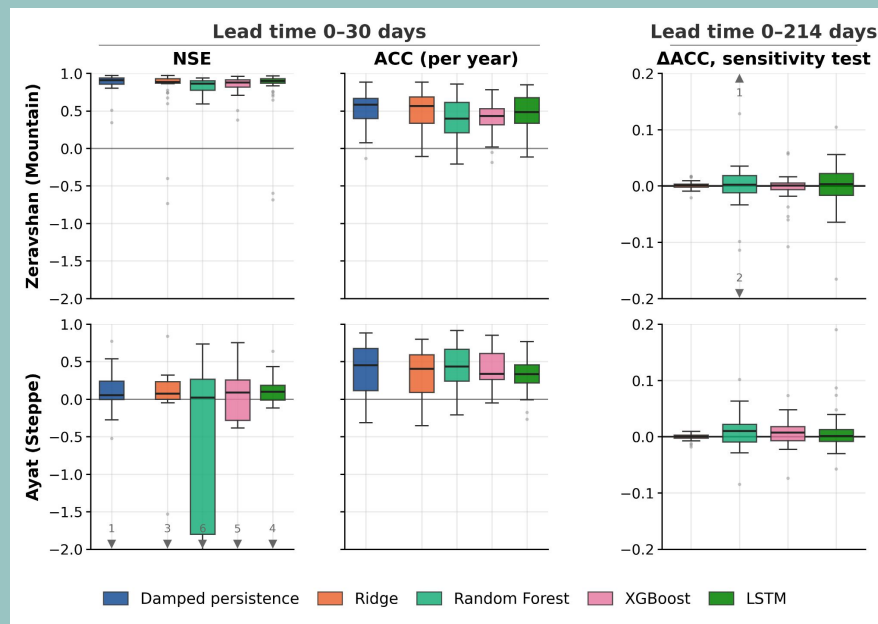


Models

- Models: Ridge, Random Forest, XGBoost, LSTM
- References: climatology, persistence, damped persistence



Results



- Mountain (Zeravshan):** Every forecast reaches high NSE, but none significantly beats damped persistence on anomaly skill
- Steppe (Ayat):** Anomaly skill comes from persistence, and volumetric accuracy collapses for all forecasts
- Sensitivity test:** Swapping SEAS5 forcing for another year's barely changes skill: models rely on antecedent discharge, not weather

Reference: Pineda Schwarz D, Didovets I, Fallah B and Tillakarim T. Can SEAS5 seasonal forecasts drive daily streamflow prediction? Machine learning experiments in Central Asian catchments. *Journal of Hydrology: Regional Studies*. 2026.

Conclusion: No learned model significantly outperformed damped persistence in the mountain catchments, and in the steppe catchments anomaly skill came from persistence while volumetric accuracy collapsed. Swapping the SEAS5 forcing barely changed skill: the models rely on antecedent discharge, so damped persistence remains the benchmark to beat.